

CPI-Surprise Trading Strategy

Methodology & Approach

Signal Construction: Calculate CPI surprise = (Actual CPI - Consensus Expectation). Generate trading signal: negative surprise → Long (+1), positive surprise → Short (-1), zero surprise → Neutral (0). Positions entered next trading day (t+1) following CPI announcement, with daily rebalancing between equity (SPY, QQQ) and risk-free assets.

Data & Period: 20 years (2004-2024, 240 CPI releases). CPI from FRED, prices via Yahoo Finance, risk-free rate from Fama-French data.

Transaction Costs: Modeled at 10 bps per position change (realistic round-trip: ~5 bps equity + 5 bps risk-free asset).

Analysis: Performance metrics (Sharpe ratio, annualized returns, volatility). Period breakdown (full sample, 2004-2019, 2020-2024). Fama-French 4-factor regression to assess alpha significance. Comparative analysis with simple long-only baseline.

1. Overall Performance (Sharpe Ratios) with Buy & Hold Benchmarks

Period	SPY Strategy	SPY (w/ TC)	SPY B&H	QQQ Strategy	QQQ (w/ TC)	QQQ B&H
Full Sample	0.229	-0.022	0.736	0.197	-0.098	0.619
2004-2019	-0.036	-0.304	0.699	0.048	-0.251	0.571
2020-2024 (t-day)	0.661	0.650	0.845	0.870	0.334	0.753

Note: Strategy without transaction costs (1st column) vs. realistic costs (2nd column, 10 bps per rebalance). B&H; = Buy & Hold benchmark (green columns). t-day = same-day entry/exit (intraday trading).

2. Why Transaction Costs Hit QQQ Harder: The Cost-to-Return Ratio

The Gap Problem: QQQ Sharpe declines from 0.197 to -0.098 (gap of -0.295), while SPY declines from 0.229 to -0.022 (gap of -0.251). Why is QQQ's decline more severe?

Root Cause: Cost as Percentage of Returns. Both strategies incur approximately 2.4% annual drag from transaction costs (~240 position changes × 10 bps). However:

- QQQ Strategy generates only +0.76% annualized return (before costs)
 - SPY Strategy generates +1.04% annualized return (before costs)
- With the same ~2.4% cost drag, QQQ's proportional impact is larger. It's equivalent to losing 3x the annual return to costs, versus SPY losing roughly 2.3x. This ratio determines the final Sharpe ratio.

Critical Insight: The strategy has an implicit minimum return threshold (~2.4% annually) before costs can be absorbed. QQQ's weaker signal barely exceeds the low returns of the 2004-2019 period, making it structurally more vulnerable to cost drag.

3. Period Analysis: 2004-2019 vs. 2020-2024

Metric	2004-2019	2020-2024 Change
SPY Annual Return	-0.15%	+4.86% ✓
SPY Sharpe	-0.036	+0.661 ✓
Volatility	Lower (4.1%)	Higher (5.7%)
Regime	Low inflation	High inflation

Interpretation: CPI as a trading signal is regime-dependent. In low-inflation environment (2004-2019), inflation surprises provide minimal trading edge and strategy loses money. In high-inflation environment (2020-2024), same signal generates strong Sharpe ratios (0.66+), indicating inflation volatility drives strategy returns.

4. Cumulative Returns on CPI Event Days (t+1 Entry)

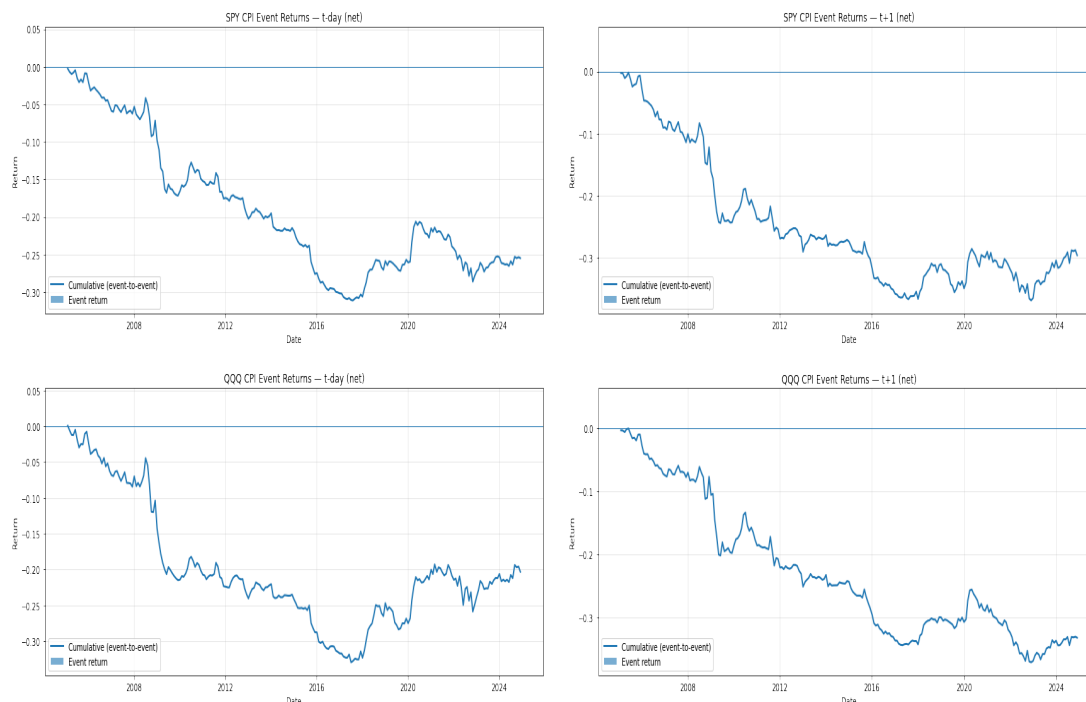


Chart Interpretation: Cumulative returns show persistent drawdown 2008-2018, recovery 2020-2024. Pre-COVID period (2004-2019) shows strategy underperformance, post-COVID shows improvement and mean reversion. The 2020-2024 inflection point coincides with elevated inflation regime, supporting regime-dependency hypothesis.

5. Comparison: Directional Signal vs. Simple Long-Only

Strategy	SPY Sharpe	QQQ Sharpe	Finding
CPI Long-Only (all CPI days)	0.726	0.762	Strong baseline
Directional Signal (Long/Short)	0.229	0.197	Weaker than long-only
Difference	-0.497	-0.565	Signal direction doesn't add value

Key Finding: Being long on ALL CPI days (0.726 Sharpe) significantly outperforms attempting to predict direction (0.229 Sharpe). This suggests CPI events themselves signal higher volatility/opportunity, but directional prediction adds noise rather than signal.

6. Factor Analysis: Fama-French 4-Factor Regression

Period	Alpha (ann)	t-stat	p-value	Significant?
Full Sample	0.75%	0.649	0.516	✗ No
2004-2019	-0.11%	-0.092	0.927	✗ No
2020-2024	2.42%	0.982	0.326	✗ No

Conclusion: No statistically significant alpha across any period (all p-values > 0.30, well above 0.05 threshold). After controlling for market, size, value, and momentum exposure, strategy returns are indistinguishable from zero. This does not disprove the CPI signal's existence, but indicates returns may reflect standard factor exposures rather than pure alpha generation.

Analysis period: 2004-2024 (240 CPI releases) | Transaction costs: 10 bps per rebalance